PrintAway | Browser-to-Printer Troubleshooting

Engineering investigation brief for Antigravity | Xerox B1025 | Windows host | 5 November 2026

Executive summary

The one-page PDF successfully prints using **scripts/test-print-proof.ps1 -SendTestPrint**, but the browser path stops after successful admin approval. The C# worker defaults to **simulateSpoolerOnly = true**. This is a strong lead, not a confirmed root cause. Worker startup, active process, printer dispatch, and queued job state remain unverified.

Observed evidence

Area	Observation
Physical test	PowerShell test sends the PDF to Xerox B1025
Web flow	Upload, quote, order, simulated payment, admin approval work
After approval	No physical paper prints
PrintWorker.cs	Constructor defaults simulateSpoolerOnly to true
Worker loop	StartBackground silently swallows exceptions with catch { }
Process check	No PrintAway.exe process appeared in prior query
Project copies	Documents\PrintAway and Downloads\PrintAway\PrintAway

Relevant code: PrintWorker.cs

```
public LocalPrintWorker(PrintAwayRepository repo,
                        bool simulateSpoolerOnly = true)

PdfAndPrinterEngine.DispatchPrintJob(
    targetQueue, orderCode, pdfBytes, pagesToPrint,
    copies, duplexMode, _simulateSpoolerOnly);
```

Do not change the constructor default to false without confirming the queue contents and runtime configuration: old approved jobs might print unexpectedly.

Investigation order - read-only first

• Identify which process owns TCP port 8080, its executable path and command line. Confirm the active project directory and database.

• Find every LocalPrintWorker constructor call and StartBackground invocation; determine whether the worker runs and its effective simulation setting.

• Trace the admin approval endpoint into job creation/release and ClaimNextQueuedJob(). Verify exactly one job is created per approved order.

• Read PdfAndPrinterEngine.DispatchPrintJob() and compare its live path with the known-working PowerShell test.

• Inspect approved, queued, claimed, failed, and reconciliation_required jobs read-only. Do not clear or replay existing jobs.

Suggested diagnostics on the Windows host

```
Get-NetTCPConnection -LocalPort 8080 -State Listen |
 ForEach-Object { Get-CimInstance Win32_Process -Filter
 "ProcessId = $($_.OwningProcess)" |
 Select ProcessId, Name, ExecutablePath, CommandLine }

Select-String -Path "C:\Users\Vikash\Documents\PrintAway\src\*.cs" `
 -Pattern "new LocalPrintWorker|StartBackground|simulateSpoolerOnly" `
 -Context 4,4

Select-String -Path "C:\Users\Vikash\Documents\PrintAway\src\PdfAndPrinterEngine.cs" `
 -Pattern "DispatchPrintJob" -Context 5,65
```

Antigravity assignment - diagnosis and implementation

• Audit application startup, worker polling, lease handling, database paths, live/simulation flags, and printer queue selection. Report confirmed cause with file/line evidence.

• Replace catch { } with structured exception logging; report worker heartbeat, selected printer, current mode, queue depth, event history and actionable error details.

• Keep simulation mode the safe default. Live mode must require an explicit operator setting and be visible in the dashboard.

• Ensure admin approval is authenticated, payment-gated and idempotent. Never duplicate jobs on repeated clicks or retries.

• Preserve recovery behavior: do not automatically reprint if spooler acceptance is uncertain. Provide manual reconciliation.

• Treat Windows spooler acceptance as submission, not proof of physical printing or pickup readiness.

• Avoid logging PDF contents, credentials, webhook secrets, or personal documents.

Validation and physical-print safety

• First run upload, quote, order, payment, approval, worker claim, PDF retrieval, rendering, and simulation dispatch tests against a disposable test database.

• Test failure handling, restart, expired leases and duplicate approval protection. Preserve production data and existing orders.

• Before enabling live mode, display all outstanding approved/queued jobs and verify the printer name and queue.

• Ask the operator before releasing any physical test. Then print exactly one approved one-page PDF and verify paper output.

Acceptance criteria and report back

• Root cause is demonstrated, not guessed, with exact source files and runtime evidence.

• Browser-approved, paid job moves to the intended queue and worker, dispatches to the correct Xerox B1025, and logs its actual spooler result.

• Simulation is clearly distinguishable from live printing; duplicate printing is prevented.

• Antigravity supplies a minimal patch, test results, exact startup command and explicit live-mode procedure.

Instruction to Antigravity: Diagnose and implement safe fixes, but do not trigger physical printing or release old queued jobs without explicit authorization.